

Effect of non-steroidal anti-inflammatory drugs and prostaglandins on the permeability of the human small intestine.

Intestinal permeability was estimated in healthy subjects after ingestion of aspirin (1.2+1.2 g), ibuprofen (400+400 mg) and indomethacin (75+50 mg) at midnight and an hour before starting a ^{51}Cr labelled ethylenediaminetetraacetate absorption test. Intestinal permeability increased significantly from control levels following each drug and the effect was related to drug potency to inhibit cyclooxygenase. Intestinal permeability increased to a similar extent after oral and rectal administration of indomethacin showing that the effect is systemically mediated. Prostaglandin E₂ decreased intestinal permeability significantly but failed to prevent the indomethacin induced increased intestinal permeability. These studies show that non-steroidal anti-inflammatory drugs disrupt the intestinal barrier function in man and suggest that the morphological correlates of the damage may reside at the level of the intercellular junctions.